

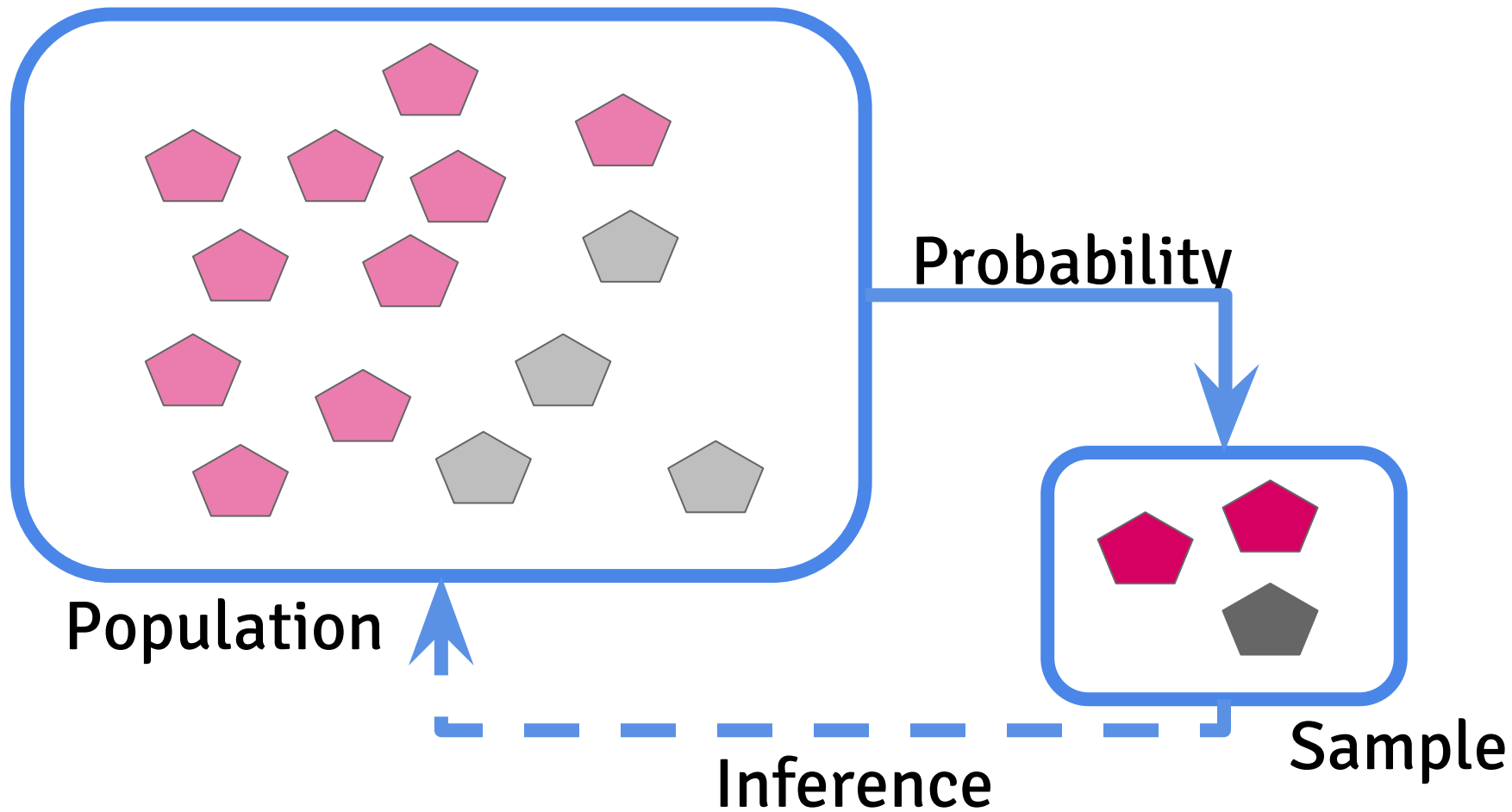
Representing data

Jeff Leek

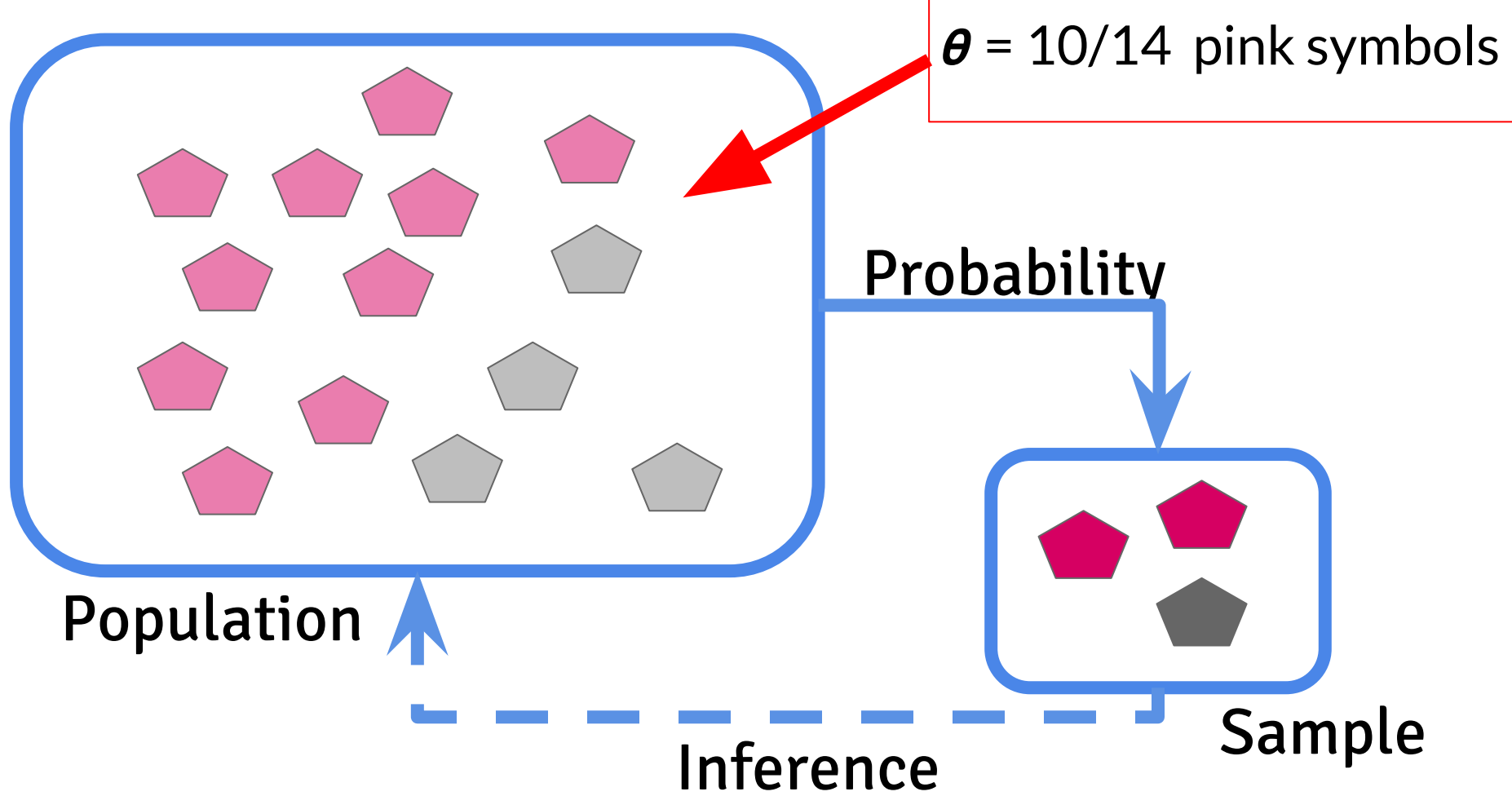
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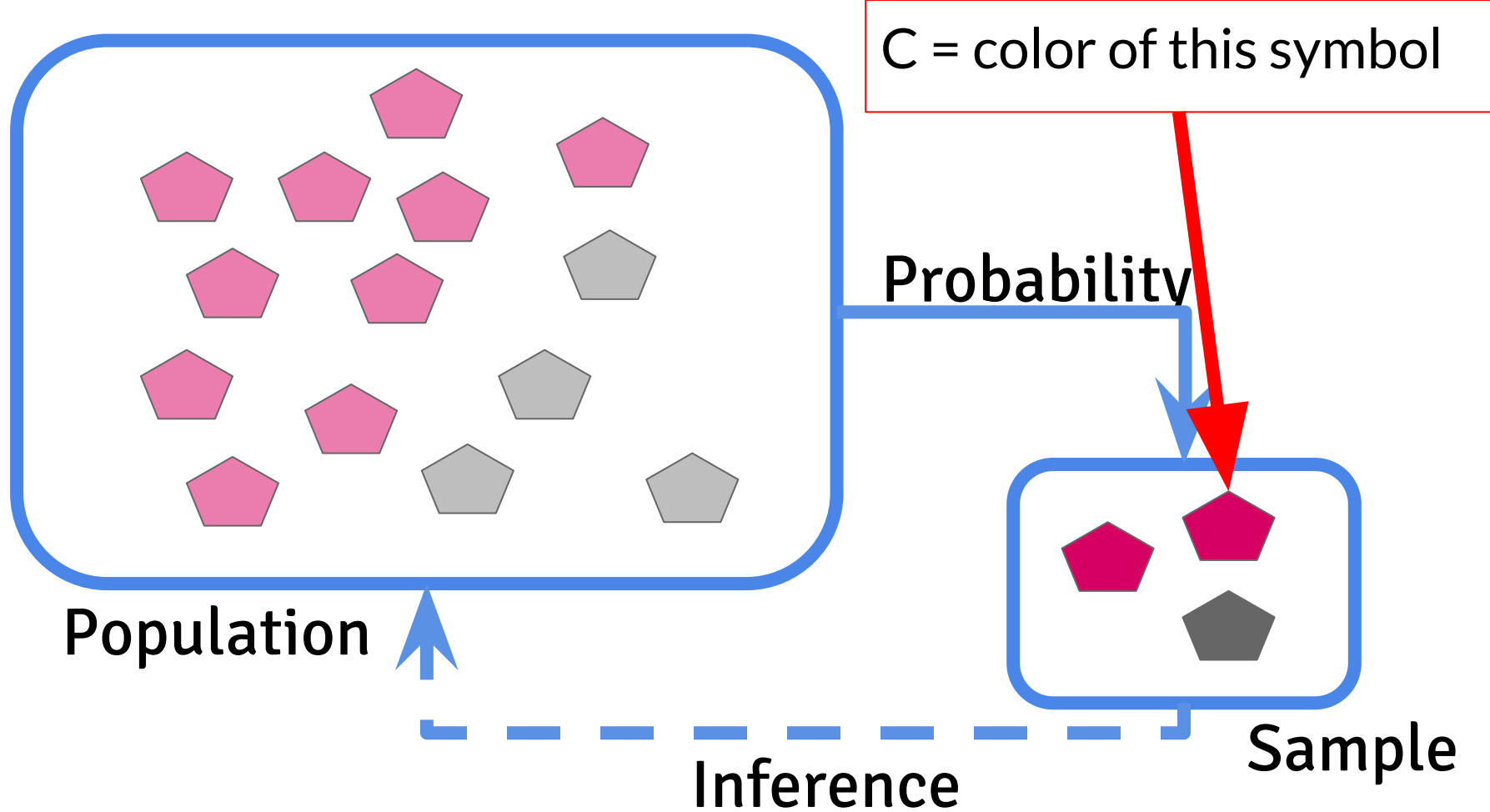
Central dogma of statistics



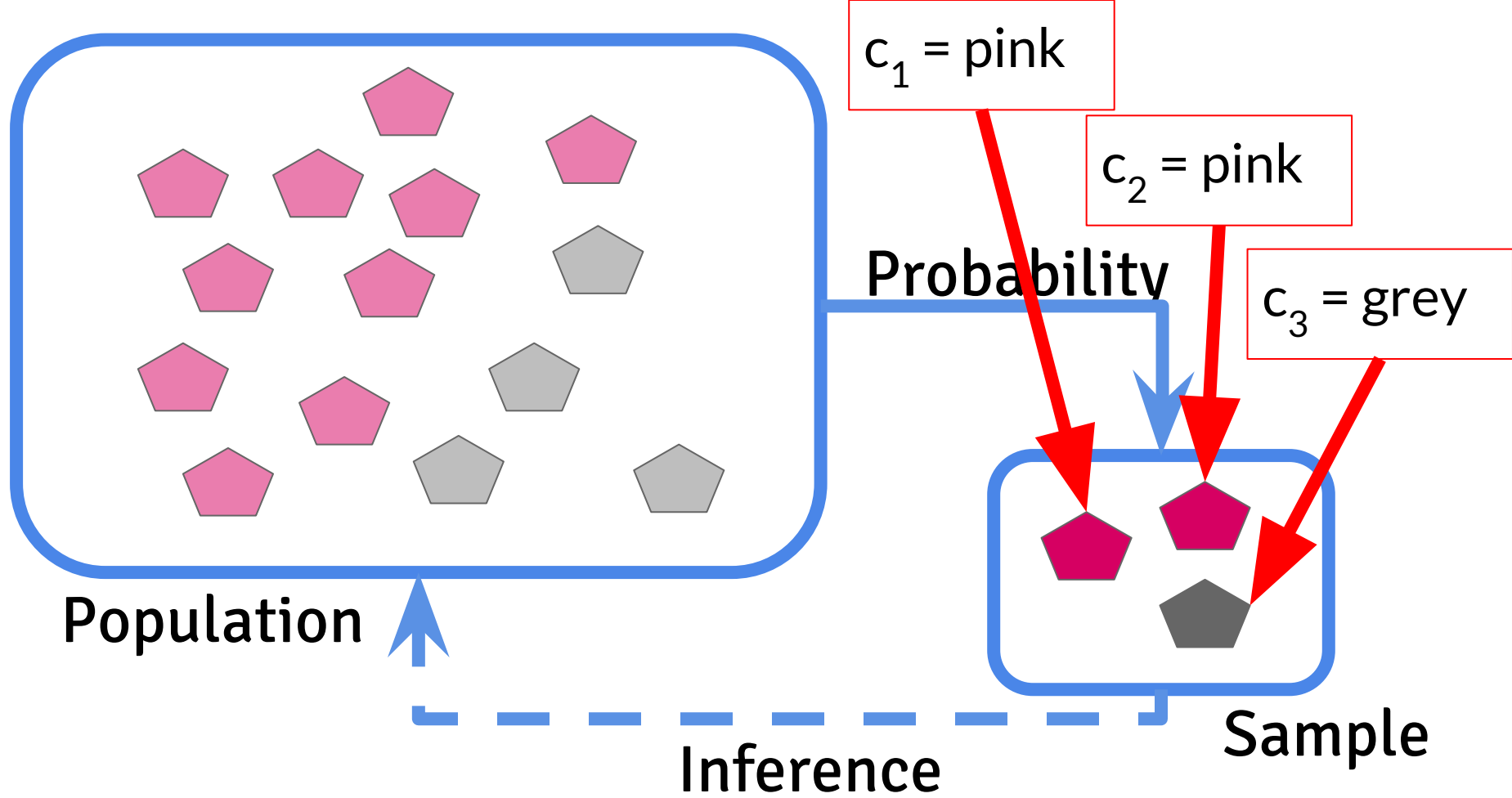
Parameters are characteristics of the population



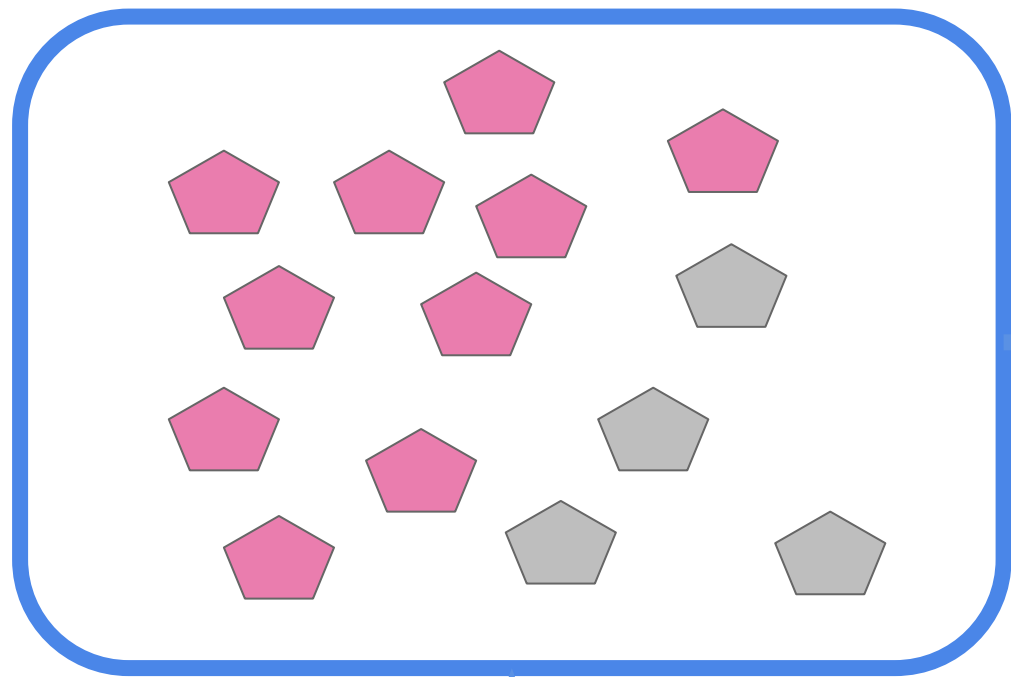
Data points are values we measure



Multiple values have subscripts



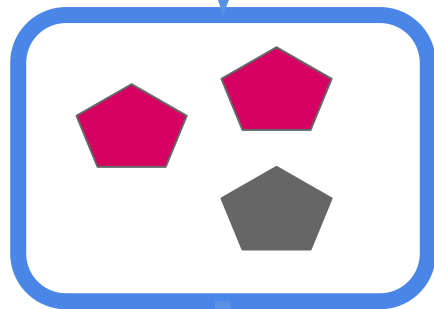
**We estimate population parameters
with the data**



Population

$$\hat{\theta} = 2/3 \text{ pink symbols}$$

Probability



Sample

Inference

Summary of notation conventions

- Data points are represented by letters
 - H for height, W for weight, C for count etc.
- Subscripts are used for different data points
 - C_1, C_2, C_3 are the counts for people 1,2,3
- Sometimes we write X for all values
 - X_1, X_2, X_3 are the counts for people 1,2,3
- We may need another subscript
 - X_{11} for the count for gene 1 on person 1

- Parameters are Greek letters
 - θ is average height in population
- Hats are used for estimates
 - $\hat{\theta}$ is our estimate of average height in population
- Y is usually outcome, X is usually covariate